

Ex.no:9	MINI PROJECT – Tourism Analytics

Aim:

To clean, transform, store, and analyze tourism data using SQL Server, Power BI, and Tableau to identify tourism trends, country-wise performance, and relationships between tourism and economic indicators.

Dataset Description:

The dataset contains tourism and economic information for different countries and years. The major attributes include country, country code, year, tourism arrivals, tourism receipts, tourism exports, tourism departures, tourism expenditures, GDP, inflation, and unemployment.

Algorithm:

1. Load the tourism dataset and inspect its structure, attributes, missing values, and data types.
2. Create a database named TourismAnalytics in SQL Server.
3. Import the tourism dataset into SQL Server as the TourismData table.
4. Check and correct the data types of the columns.
5. Connect Power BI to the SQL Server database and select the TourismData table.
6. Clean and transform the data using Power Query by checking missing values, duplicates, and data types.
7. Load the transformed data into Power BI.
8. Create DAX measures for total tourist arrivals, tourism receipts, tourism expenditure, tourism exports, average tourist arrivals, and average unemployment.
9. Create KPI cards to display important tourism metrics.
10. Create visualizations such as line charts, bar charts, and treemaps to analyze tourism trends and country-wise performance.
11. Add interactive filters and slicers such as the Year slicer.
12. Apply filtering and drill-down or drill-through features where required.
13. Connect the dataset to Tableau and prepare the data for visualization.
14. Create calculated fields, charts, filters, parameters, dashboards, and stories in Tableau.
15. Analyze the visualizations to identify tourism trends and country-wise performance.
16. Present the final tourism analysis using the Power BI dashboard.

Output:

Installation has completed successfully!

INSTANCE NAME	CONNECTION STRING
MSSQLSERVER	Server=localhost;Database=master;Trusted_Connection=True;
SQL ADMINISTRATORS	SQL SERVER INSTALL LOG FOLDER
AAFRIN\91754	C:\Program Files\Microsoft SQL Server\170\Setup Bootstrap\Log\2026091
FEATURES INSTALLED	INSTALLATION MEDIA FOLDER
SQLENGINE	C:\SQL2025\StdDev_ENU
VERSION	INSTALLATION RESOURCES FOLDER
17.0.1000.7, RTM	C:\Program Files\Microsoft SQL Server\170\SSE\Resources

Connect

History Browse

Type here to filter the list

- ✓ Favorites
No favorite connections
- ✓ Recent Connections
No recent connections

Connection Properties

Connection String

Server Name	localhost
Authentication	Windows Authentication
User Name	AAFRIN\91754
Password	
	☐ Remember Password
Database Name	<default>
Encrypt	Mandatory
	☒ Trust Server Certificate
Host Name In Certificate	

Advanced...

Custom Properties

Name	
Color	<default>

Reset

Connect

Cancel

Preview Data

This operation analyzed the input file structure to generate the preview below for up to the first 50 rows.

country	country_code	year	tourism_receipts	tourism_arrivals	
Aruba	ABW	1999	782000000.0	972000.0	6
Africa Eastern a...	AFE	1999	8034209108.0477	15309377.8420...	1
Afghanistan	AFG	1999			
Africa Western ...	AFW	1999	1443612846.65...	3897975.03054...	3
Angola	AGO	1999	31000000.0	45000.0	0
Albania	ALB	1999	218000000.0	371000.0	5
Andorra	AND	1999		9422000.0	
Arab World	ARB	1999			
United Arab Em...	ARE	1999	893000000.0		
Argentina	ARG	1999	3175000000.0	2898000.0	1
Armenia	ARM	1999	47000000.0	41000.0	1
American Samoa	ASM	1999		41000.0	
Antigua and Ba...	ATG	1999		560000.0	
Australia	AUS	1999	11408000000.0	4459000.0	1
Austria	AUT	1999	12358000000.0	17467000.0	
Azerbaijan	AZE	1999	93000000.0	602000.0	7
Burundi	BDI	1999	1200000.04768...	26000.0	2
Belgium	BEL	1999		6369000.0	
Benin	BEN	1999	94400001.5258...	941000.0	1
Burkina Faso	BFA	1999		117000.0	

☒ Use Rich Data Type Detection - may provide a closer type fit. However, cells with anomalous values may be dropped.

Modify Columns

This operation generated the following table schema. Please verify if schema is accurate, and if not, please make any changes.

Column Name	Data Type	Primary Key	☐ Allow Nulls	
country	nvarchar(100)	☐	☐	
country_code	nvarchar(50)	☐	☐	
year	int	☐	☐	
tourism_receipts	float	☐	☒	
tourism_arrivals	float	☐	☒	
tourism_exports	float	☐	☒	
tourism_departures	nvarchar(50)	☐	☒	
tourism_expenditures	float	☐	☒	
gdp	float	☐	☒	
inflation	float	☐	☒	
unemployment	nvarchar(50)	☐	☒	

Import Flat File 'TourismAnalytics'

Results

Introduction

Specify Input File

Preview Data

Modify Columns

Summary

Results

Operation Complete

Summary:

Name	Result
Insert Data	Success

SQLQuery1...91754 (72)j*

```
1 SELECT *
2 FROM dbo.TourismData;
3
4 SELECT COUNT(*) AS Total_Rows
5 FROM dbo.TourismData;
```

Results Messages

country	country_code	year	tourism_receipts	tourism_arrivals	tourism_exports	tourism_departures	tourism_expenditures	gdp	inflation	unemployment
Middle income	MIC	2002	NULL	394155191.535862	8.53367603993628	318157017.889631	6.04231291858411	9871238528925.74	4.26870632309104	NULL
North Macedonia	MKD	2002	5000000	123000	5.95266234482291	NULL	3.2691722389795	401236701.08972	2.31499794662219	31.84
Mali	MLI	2002	10500000	96000	10.0051219025309	NULL	5.81254486534784	3908121307.01957	5.0328227571195	NULL
Malta	MLT	2002	75700000	1483000	21.100372861813	NULL	5.17299837718984	4455128322.74822	2.18849404005049	6.902
Myanmar	MMR	2002	13800000	457000	5.81429296252378	NULL	1.77801631832048	67783252.0701	57.0245113561713	NULL
Middle East & North Africa	MNA	2002	2030885241.1459	29083175.7213185	20.4090876174054	NULL	6.43771158827318	483182700952.368	2.7291356925594	13.69623803
Montenegro	MNE	2002	NULL	136000	NULL	NULL	NULL	128480255.52413	NULL	6.2
Mongolia	MNG	2002	14300000	239000	20.183488992081	NULL	13.241897075247	139550579.87409	5.917074687184271	NULL
Northern Mariana Islands	MNP	2002	NULL	470000	NULL	NULL	NULL	128400000	NULL	NULL
Mozambique	MZQ	2002	6800000	943000	NULL	NULL	NULL	5992789428.80213	NULL	NULL
Mauritania	MRT	2002	NULL	NULL	NULL	NULL	NULL	1777851461.26043	3.8957052570849	NULL
Total_Rows										
1			6850							

Untitled - Power Query Editor

Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Text Use First Row as Headers Replace Values Merge Append

Queries [1]

Source[["Schema="dbo",Item="TourismData"]][Data]

country	country_code	year	1.2 tourism_receipts	1.2 tourism_arrivals	1.2 tourism
Aruba	ABW	1999	782000000	972000	
Africa Eastern and Southern	AFE	1999	8034209108	15309377.84	
Afghanistan	AFG	1999	null	null	
Africa Western and Central	AFW	1999	1443612847	3897975.031	
Angola	AGO	1999	31000000	45000	
Albania	ALB	1999	218000000	371000	
Andorra	AND	1999	null	9422000	
Arab World	ARB	1999	null	null	
United Arab Emirates	ARE	1999	893000000	null	
Argentina	ARG	1999	3175000000	2898000	
Armenia	ARM	1999	47000000	41000	
American Samoa	ASM	1999	null	41000	
Antigua and Barbuda	ATG	1999	null	560000	
Australia	AUS	1999	11408000000	4459000	
Austria	AUT	1999	12358000000	17467000	
Azerbaijan	AZE	1999	93000000	602000	
Burundi	BDI	1999	1200000.048	26000	
Belgium	BEL	1999	null	6369000	

11 COLUMNS. 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 13:27

Query Settings

PROPERTIES

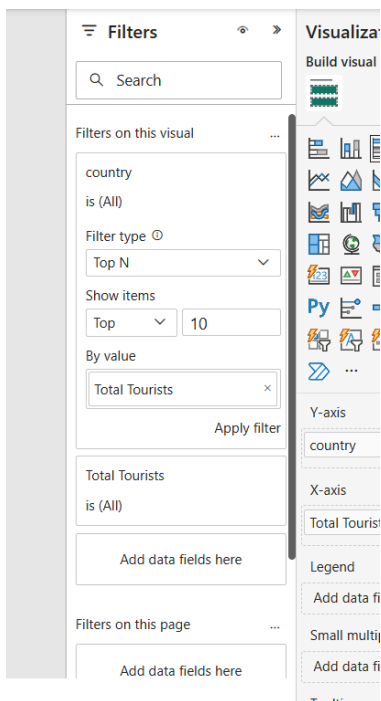
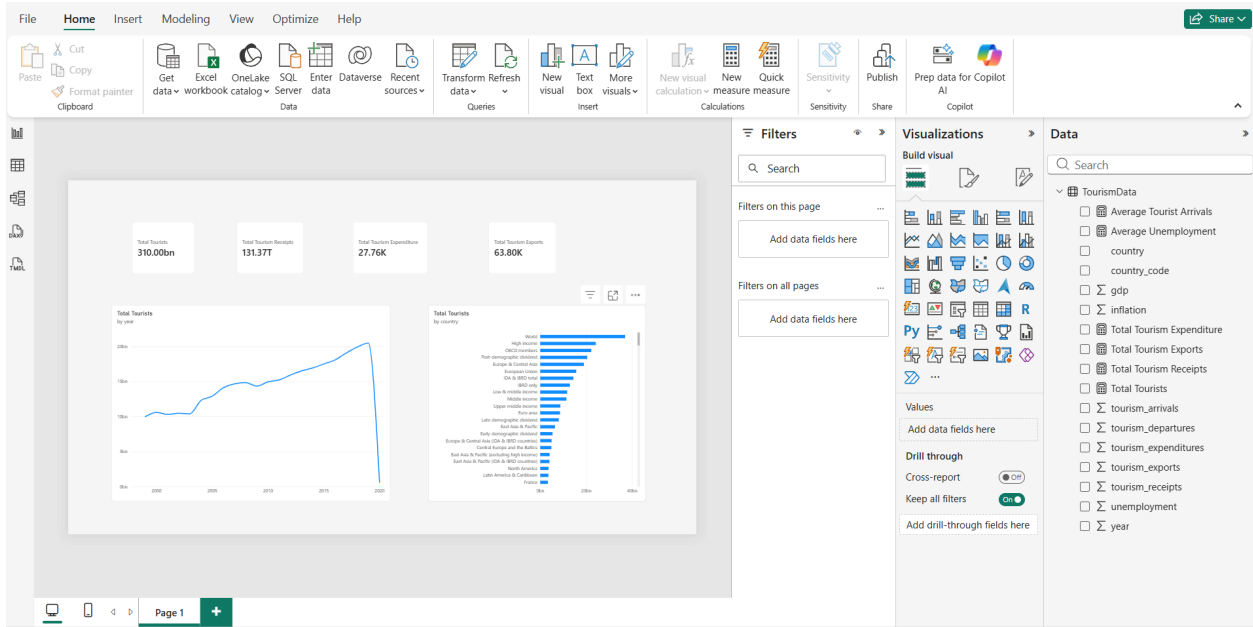
Name

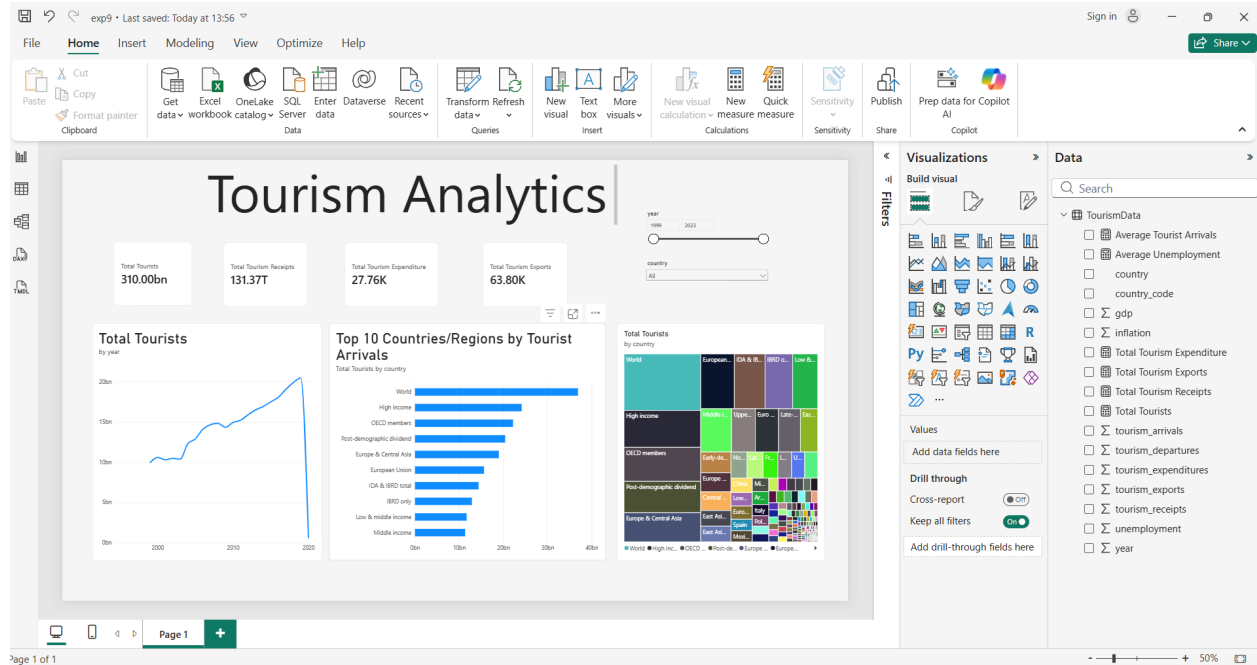
TourismData

APPLIED STEPS

Source

Navigation





Interpretation:

The KPI cards provide an overview of the major tourism indicators. The line chart shows changes in tourist arrivals over different years. The bar chart allows comparison of tourist arrivals across countries or regions. The treemap provides a visual comparison of tourism arrivals. The Year slicer allows the user to interactively analyze tourism performance for selected years.

Result:

The tourism dataset was successfully stored in SQL Server, connected to Power BI, cleaned, and analyzed using DAX measures and interactive visualizations. The Power BI dashboard provides an effective way to analyze tourist arrivals, tourism receipts, expenditure, exports, and year-wise tourism trends.